

CONNECT WITH PLYMOUTH UNIVERSITY

ELEC 345

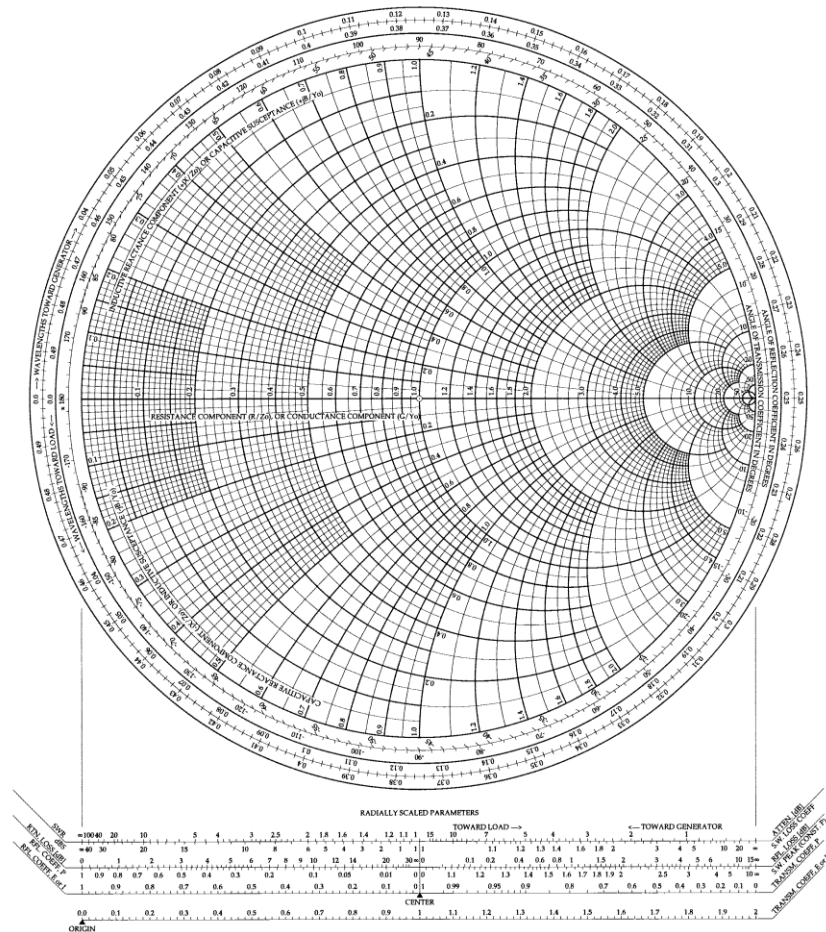
High speed communications

Stub matching

Last week:

1. We looked at transmission lines.
2. Loss, radiation, propagation.
3. The wave equation
4. Matched loads
5. Reflections
6. We've reminded ourselves how the smith chart works and looked at a few L matches.

The Smith Chart



Basic tools:

1. Series (impedance, Z) values:

 = -j down the smith chart  = +j up the smith chart

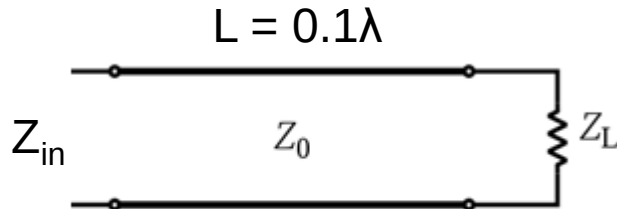
2. Shunt Y (parallel) values:

 = +j up the smith chart  -j down the smith chart

Normalized to centre of the chart, divide by characteristic impedance,
 $1 = 50\Omega$ or 75Ω or N where N = characteristic impedance etc. etc.

Z to Y & vice versa mirror though the centre of the chart.

Finding Z_L given Z_{in}

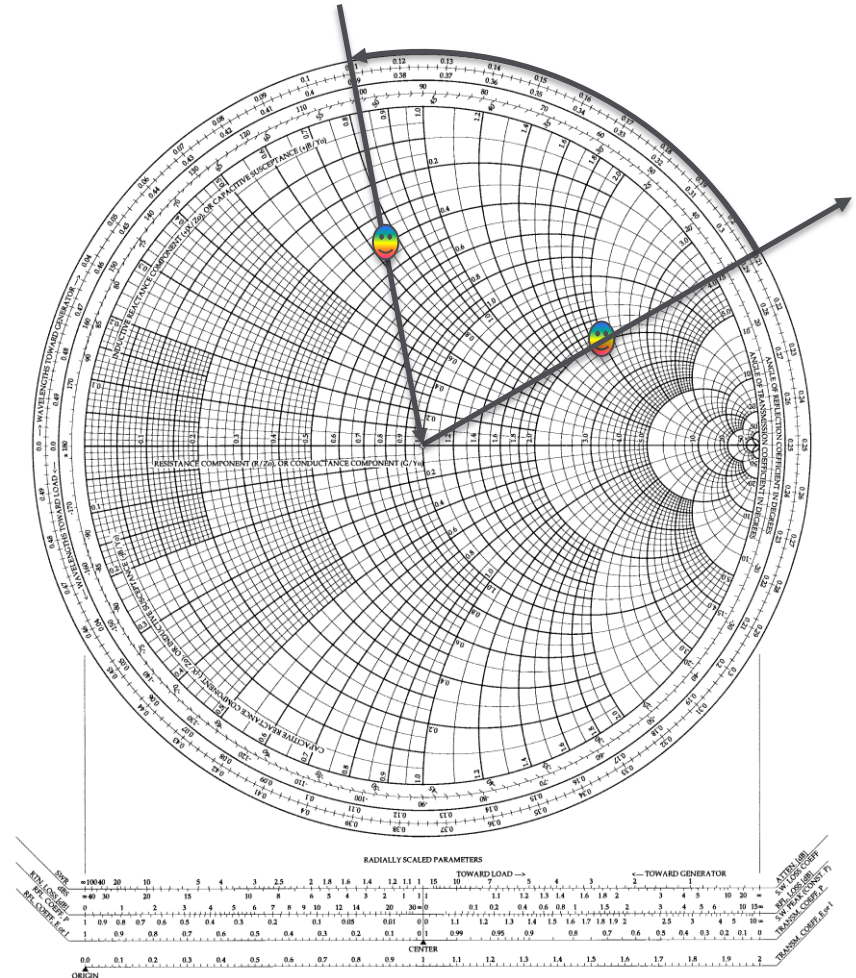


$$Z_{in} = 100 + j100\Omega$$

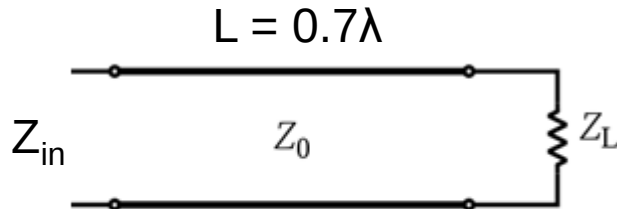
- Normalize the value to plot on the smith chart (divide by 50Ω)

$$z = 2 + j2$$

- Take a line from the centre to the circumference
- Note the value on the WTL circle
- Rotate 0.1λ towards the load.
- Draw line in to centre.
- Measure correct distance and mark.
- Read off value, $0.4 + j0.75$
- Finally:



Finding Z_L given Z_{in}

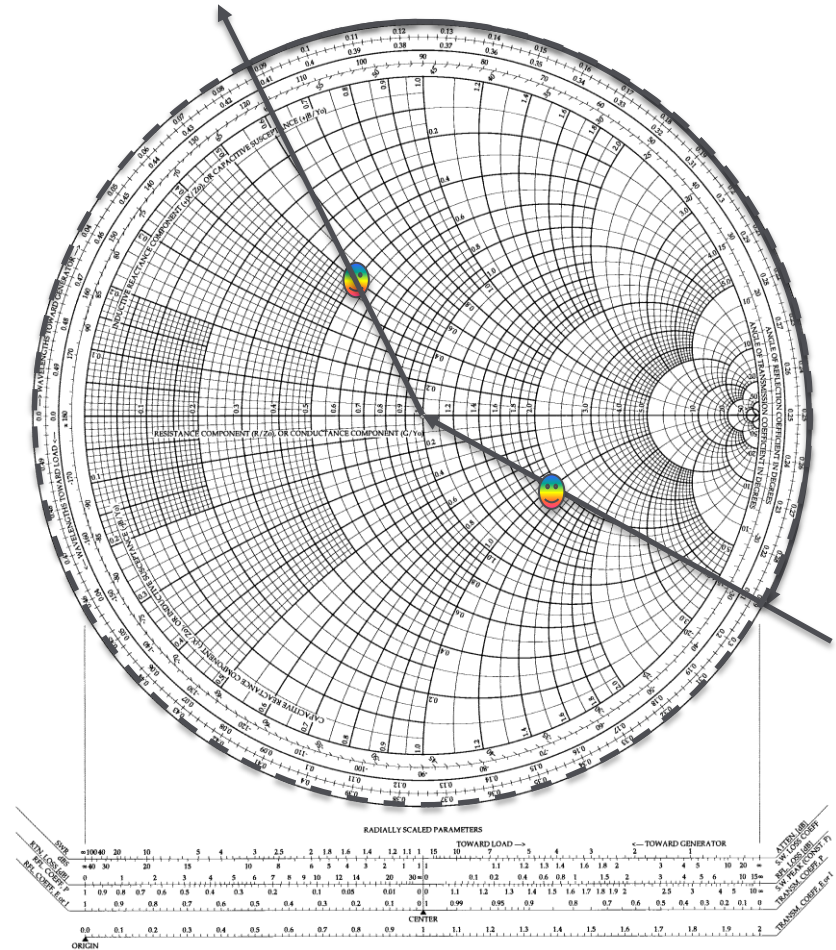


$$Z_L = 25 + j25\Omega$$

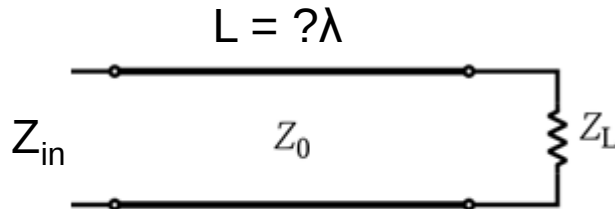
- Normalize the value to plot on the smith chart (divide by 50Ω)

$$z = 0.5 + j0.5$$

- Take a line from the centre to the circumference
- Note the value on the WTG circle
- Rotate 0.7λ towards the generator.
- Draw line in to centre.
- Measure correct distance and mark.
- Read off value, $1.9 - j$
- Finally:



Finding $Z_{in} = j0.7$ for a short circuit



$Z_L = 0$, (short circuit)

- Normalize the value to plot on the smith chart (divide by 50Ω)

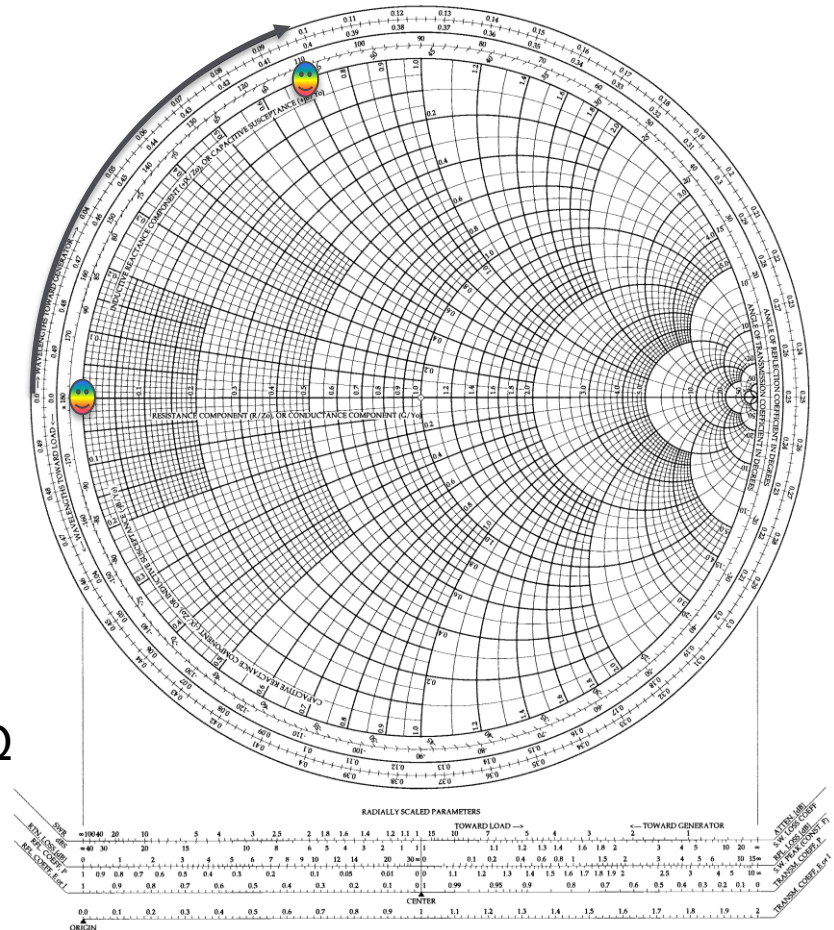
$Z_{sc} = 0$

$Z_{in} = j35\Omega$

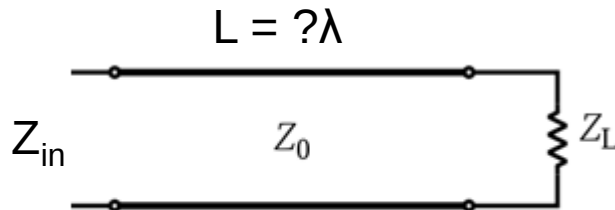
- Normalize the value to plot on the smith chart (divide by 50Ω)

$Z_{\lambda} = j0.7$

- Note the value on the WTG circle
- Measure how far you need to rotate the short circuit to reach this point.
- Short circuit stub, $L = 0.098\lambda \Rightarrow 0 + j35\Omega$



Finding $Z_{in} = -j$ for an open circuit



$Z_L = \infty$, (open circuit)

- Normalize the value to plot on the smith chart (divide by 50Ω)

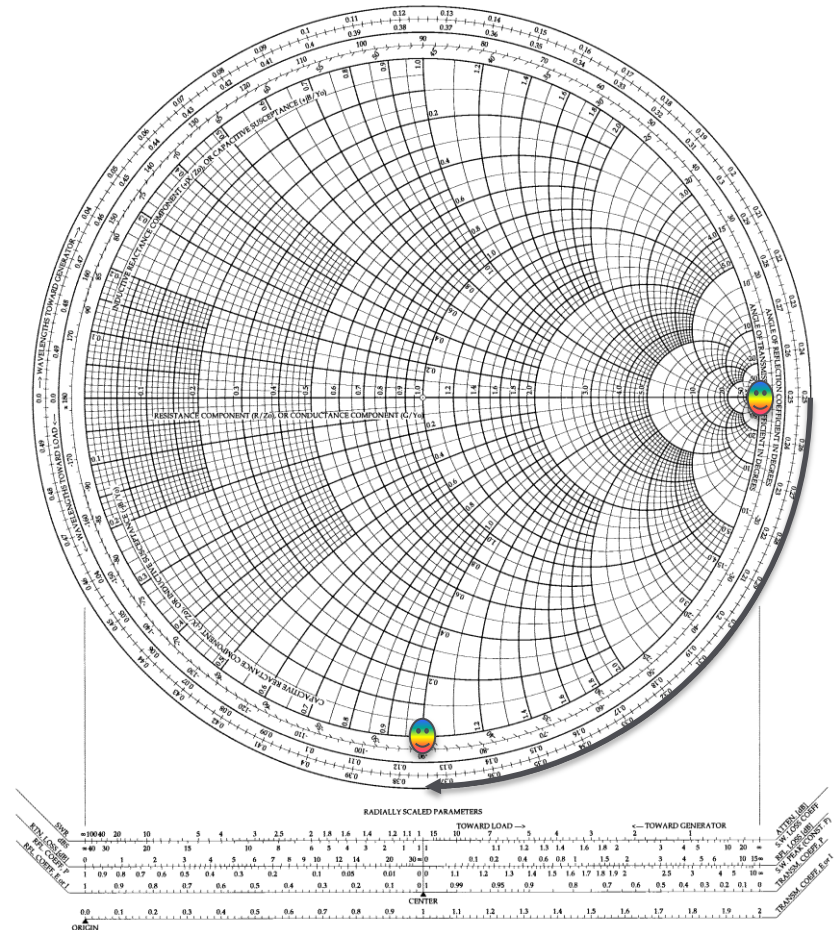
$Z_{oc} = \infty$

$Z_{in} = -j50\Omega$

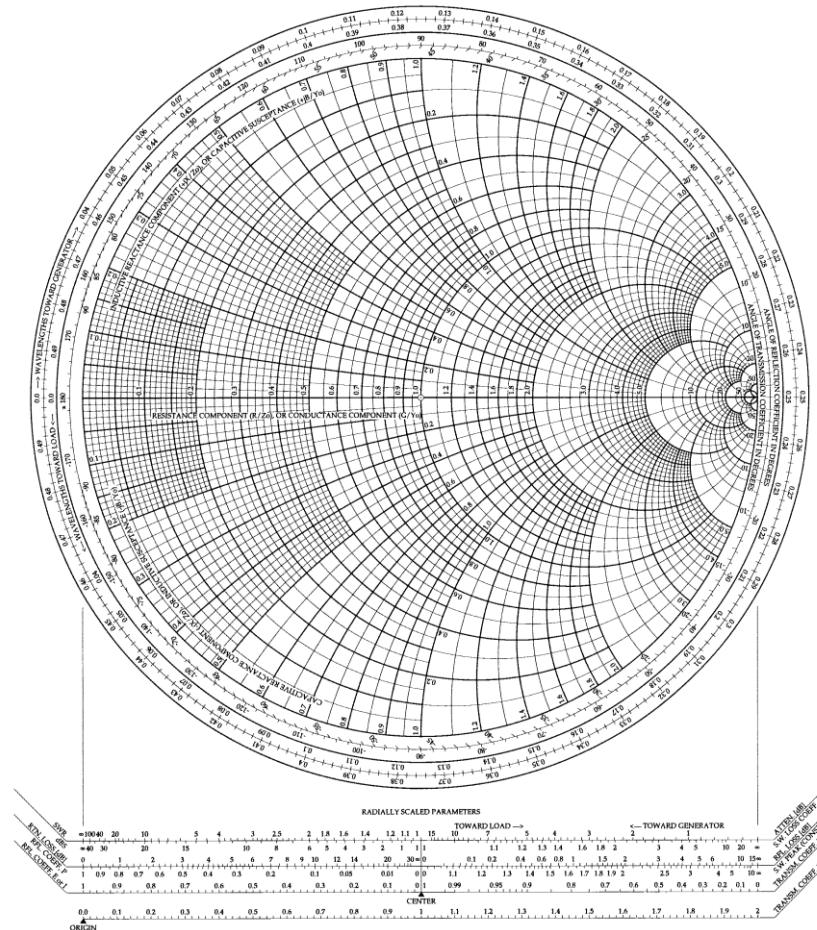
- Normalize the value to plot on the smith chart (divide by 50Ω)

$Z_{\lambda} = -j$

- Note the value on the WTG circle
- Measure how far you need to rotate the open circuit to reach this point.
- Open circuit stub, $L = 0.125\lambda \Rightarrow 0 + j50\Omega$



Putting it all together



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